DINESH NATARAJAN

Aerospace Engineer | FEA · CFD · Flight Dynamics · GD&T · V&V

dineshnatarajan02@gmail.com | +91 80985 74450
linkedin.com/in/dinesh-natarajan-baaa95206
Bengaluru, India

Aircraft Engine Bracket — GD&T & Tolerance Analysis

ASME Y14.5-2018 · A|B|C datum scheme · worst-case + RSS stack-up · MMC fastener fit · design-for-manufacture extension

SUMMARY

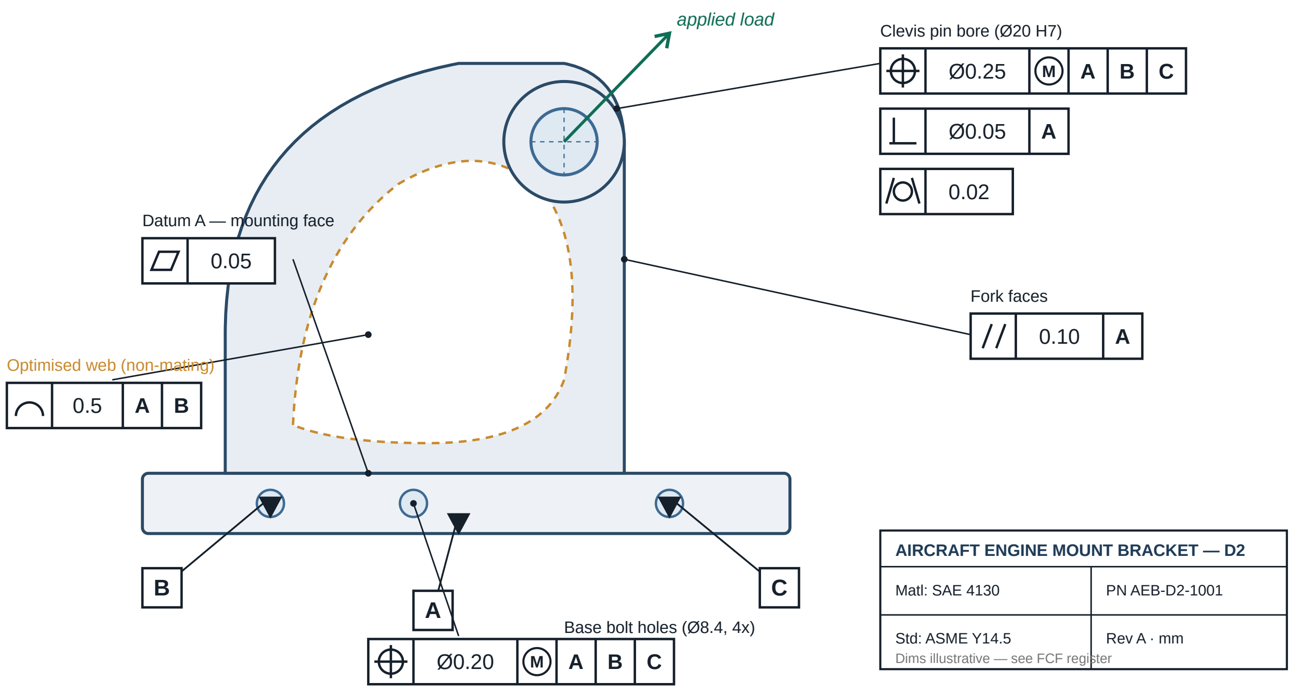
Extended the validated topology-optimised engine mount bracket (Design 2, SAE 4130) into a fully manufacturable, inspectable definition. Assigned a function-driven A|B|C datum reference frame and specified geometric controls only on mating and load-path features, leaving the non-mating optimised web to a deliberately loose profile tolerance. Verified the bolted interface with a 1D tolerance stack-up (worst-case and statistical RSS) and a floating-fastener / MMC analysis — proving the assembly fits at the tolerance extremes. Demonstrates closed-loop V-Model thinking: analysis → optimisation → producible, tolerance-controlled design.

KEY RESULTS

Governing standard	**ASME Y14.5-2018**
Datum reference frame	**A | B | C (3-2-1, fully constrained)**
Geometric controls	**10 — function-driven, not blanket-tight**
Bore-height stack (worst-case)	**± 0.35 mm**
Bore-height stack (RSS)	**± 0.206 mm (58.9% of worst-case)**
Base-hole position	**Ø0.20 Ⓜ — floating-fastener derived**
Allowable position at LMC	**Ø0.40 (incl. MMC bonus)**
Assembly guarantee	**✓ Proven via virtual condition**

METHODOLOGY

1. **Datum reference frame** — A = base mounting face (3 DOF); B, C = base locating holes (2+1 DOF). Function-driven 3-2-1.
2. **Feature classification** — Separate mating / load-path features from the non-mating topology-optimised web.
3. **FCF assignment** — Form, orientation and location controls applied per ASME Y14.5-2018 only where justified.
4. **1D tolerance stack-up** — Pin-bore height above datum A — worst-case (arithmetic) and RSS (statistical) bands.
5. **Fastener-fit / MMC** — Floating-fastener rule sets Ø0.20 Ⓜ; bonus + virtual condition prove assembly.
6. **Inspection plan** — CMM datum fit, bore cylindricity scans, GD&T profile (3D scan vs CAD) on the web.
7. **Deliverables** — Annotated 2D GD&T drawing + multi-sheet tolerance stack-up workbook.



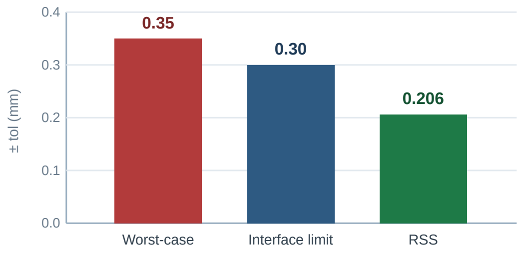
GD&T-annotated side elevation — datum scheme (A|B|C) and feature control frames per ASME Y14.5-2018

TOLERANCE STACK-UP (PIN-BORE HEIGHT)

FASTENER FIT & MMC BONUS

Floating fastener: position Ø0.20 = (hole MMC 8.40 − screw MMC 8.00) / 2 per part.

MMC bonus: at LMC (Ø8.60), bonus Ø0.20 → total allowable position **Ø0.40**.



RSS band lands below the mating-clevis allowance — interface will not bind.

Virtual condition: 8.40 − 0.20 = **Ø8.20** > screw MMC 8.00 → ✓ **assembly guaranteed** at every size within limits.

TOOLS CATIA V5 / SolidWorks (GD&T drawing) | ASME Y14.5-2018 | Tolerance stack-up (Excel) | CMM / 3D-scan inspection planning

RELEVANCE Design-for-Manufacture | Dimensional management | Tolerance stack-up (WC / RSS) | Datum schemes & GD&T | Geometric inspection

Applicable to roles at Airbus, Safran, GE Aerospace, Collins Aerospace, GKN and aerospace ESPs.